

L12: Generative Deep Learning

CSci 560 Deep Learning w/ Python (Chollet) Ch. 12

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L12.1 Text Generation

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How do you Generate Sequence Data?

- Train a model (usually Transformer or RNN) to predict next token
 - e.g. like the encoder / decoder architecture, though we only have 1 sequence so only need decoder
- Any network that can model the probability of the next token given the previous ones is called a **language model**.
- A language model captures the **latent space** of language: its statistical structure.
- **Sample** from a language model: feed it an initial string (**conditioning data**) ask it to generate next token.
 - Add the generated output back to the input data, repeat

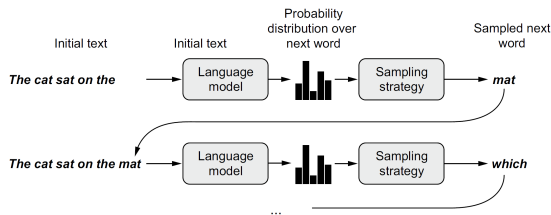


Figure 1: The process of word-by-word text generation using a language model. (Chollet, 2021, pg.367)

The Importance of the Sampling Strategy

- **Greedy sampling** consists of always choosing the most likely next token.
- **Stochastic sampling** sampling from the probability distribution for the next character.
- Useful to **control the amount of randomness** in the sampling process **softmax temperature**

```
def reweight_distribution(original_distribution, temperature=0.5):  
    """Given a softmax probability distribution, reweight the probabilities  
    using a temperature variable. Temperature 0 = greedy sampling, temperature  
    1 = uniform like (completely random) distribution  
  
    Parameters  
    -----  
    original_distribution : ndarray  
        Vector of probability over a vocabulary, probability must sum to 1.0  
    temperature : float  
        Factor quantifying the entropy of the output distribution  
  
    Returns  
    -----  
    new_distribution : ndarray  
        A reweighted version of the original distribution.  
    """  
    distribution = np.log(original_distribution) / temperature  
    distribution = np.exp(distribution)  
    # sum may longer be 1.0, so divide by the sum to obtain new probability  
    # distribution that sums to 1.0  
    return distribution / np.sum(distribution)
```



Summary

- You can generate discrete sequence data by training a model to predict the next token(s), given previous tokens.
- In the case of text, such a model is called a **language model**. It can be based on either words or characters.
- Sampling the next token requires a balance between adhering to what the model judges likely, and introducing randomness.
- One way to handle this is the notion of softmax temperature. Always experiment with different temperatures to find the right one.
- Don't expect to generate meaningful text (even for a chat GPT level of training data), all you are doing is sampling data from a statistical model of which words come after which words. Language models are all form and no substance.



L12.2 DeepDream

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DeepDream

- DeepDream is an artistic image-modification technique that uses the representations learned by convolutional neural networks.
- The DeepDream algorithm is almost identical to the convnet filter-visualization technique introduced previously consisting of running a convnet in reverse:
 - doing gradient ascent on the input to the convnet in order to maximize the activation of a specific filter in an upper layer of the convnet.

DeepDream uses this same idea, with a few simple differences:

- With DeepDream, you try to maximize the activation of entire layers rather than that of a specific filter, thus mixing together visualizations of large numbers of features at once.
- You start not from blank, slightly noisy input, but rather from an existing image—thus the resulting effects latch on to preexisting visual patterns, distorting elements of the image in a somewhat artistic fashion.
- The input images are processed at different scales (called octaves), which improves the quality of the visualizations.

- DeepDream consists of running a convnet in reverse to generate inputs based on the representations learned by the network.
- The results produced are fun and somewhat similar to the visual artifacts induced in humans by the disruption of the visual cortex via psychedelics.
- Note that the process isn't specific to image models or even to convnets. It can be done for speech, music, and more.

L12.3 Neural Style Transfer

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Neural Style Transfer



Figure 2: A style transfer example. (Chollet, [2021](#), pg.383)

- Another major development in deep-learning image modification **neural style transfer**.
- **Style** essentially means texture, colors and visual patterns.
- The key notion behind implementing style transfer is the same idea that's central to all deep learning algorithms: you define a loss function to specify what you want to achieve and you minimize this loss.

Neural Style Transfer

We know what we want to achieve: conserving the content of the original image while adopting the style of the reference image.

If we were able to mathematically define **content** and **style** than an appropriate loss function to minimize would be:

$$\text{loss} = (\text{distance}(\text{style}(\text{reference_image}) - \text{style}(\text{combination_image})) + \text{distance}(\text{content}(\text{original_image}) - \text{content}(\text{combination_image})))$$

A fundamental observation was that deep convolutional neural networks offer a way to mathematically define the style and content functions

Content Loss and Style Loss

Activations of different layers provide decomposition of contents over different spatial scales.

Content Loss

Expect content of image (global and abstract) to be captured by upper layers.
∴ A good candidate for content loss is L2 norm between activations of an upper layer in a pretrained convnet, computed over target and generated image.

Style Loss

[height=1.0in] Content uses an upper layer, but style involves all spatial scales.
∴ For style, use the inner product of the feature maps of selected layers.

You can use a pretrained convnet to define a loss that will:

- Preserve content by maintaining similar high-level layer activations between the original image and the generated image.
- Preserve style by maintaining similar correlations within activations for both low level layers and high-level layers.

L12.4 Generating Images with Variational Autoencoders

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L12.5 Introduction to Generative Adversarial Networks

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Chollet, F. (2021). *Deep learning with python* (second). Manning.